A real estate company wants to develop a system that **predicts house prices** based on square footage, number of bedrooms, and location.
Q: **Identify the problem type** and outline the **step-by-step logic** to solve it.

Answer:

Identify the problem type: **Regression**

Step-by-step logic:

1. Collect Data – **Gather historical data** with features like square footage, number of bedrooms, and location.

2. Preprocess Data – **Handle missing values**, encode categorical variables (e.g., location).

3. Split Dataset – **Divide the dataset into training and testing sets.**

4. Choose Algorithm – Use a regression model like **Linear Regression or Decision Tree Regression.**

5. Train the Model – **Fit the model on the training dataset.**

6. Evaluate Performance – Use metrics like **RMSE (Root Mean Square Error) and R^2 Score.**

7. Make Predictions – Use the model to **predict house prices for new data**.

==

A bank wants to build a model to **detect fraudulent transactions** by analyzing customer spending behavior and transaction history.
Q: **Identify the problem type** and outline **the step-by-step logic** to solve it.

Answer:

Identify the problem type: **Classification**

Step-by-step logic:

1. Collect Data – **Gather transaction records** labeled as fraudulent or non-fraudulent.

2. Preprocess Data – **Remove outliers, normalize transaction amounts**, and encode

categorical features.

3. Feature Engineering – Create features like transaction frequency, **average spending, and unusual behavior detection.**

4. Split Dataset – **Divide data into training and testing sets.**

5. Choose Algorithm – Use classification models like **Logistic Regression, Random Forest, or Neural Networks.**

6. Train the Model **– Fit the model using labeled transaction data.**

7. Evaluate Performance – Use metrics like **accuracy, precision, recall, and F1-score.**

8. Deploy Model – **Implement real-time fraud detection.**

===

A supermarket wants to segment its **customers based on their shopping patterns** to provide personalized promotions.
Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Identify the problem type: **Clustering**

Step-by-step logic:

1. Collect Data – **Gather customer purchase history**, amount spent, and frequency of purchases.

2. Preprocess Data – **Normalize data** (e.g., scale spending amounts to avoid bias).

3. Choose Clustering Algorithm – Use **K-Means, DBSCAN, or Hierarchical Clustering**.

4. Determine Optimal Clusters **– Use the Elbow Method** to find the best number of clusters.

5. Train Model – **Apply clustering algorithm** to group customers.

6. Analyze Clusters – Interpret results t**o identify high-spending, medium-spending, and low-spending customer groups.**

7. Use Clusters for Marketing – **Target each segment with personalized promotions**.

==

A **company wants to estimate an employee's salary based** on their **years of experience, job title, and education level.**
Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Identify the problem type: **Regression**

Step-by-step logic:

1. Collect Data – **Gather employee records** with years of experience, education, and salary.

2. Preprocess Data – **Handle missing values** and encode categorical variables (e.g., job title).

3. Split Dataset – **Separate data into training and testing sets**.

4. Choose Algorithm – Use **Linear Regression or Random Forest Regression.**

5. Train the Model – **Fit the model on training data.**

6. Evaluate Model – Use **Mean Absolute Error (MAE) and R^2 score** for accuracy measurement.

7. Make Predictions – **Predict salary based on new employee data**.

==

An **email provider wants** to automatically **classify incoming emails as spam or not spam** based on their content and sender details.
Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Identify the problem type: **Classification**

Step-by-step logic:

1. Collect Data – **Use datasets of spam and non-spam emails**.

2. Preprocess Data – **Convert email text to numerical format** using TF-IDF or word embeddings.

3. Split Dataset – **Divide data into training and testing sets.**

4. Choose Algorithm – Use **Naive Bayes, Support Vector Machines, or Neural Networks**.

5. Train the Model – **Fit the model using labeled email data**.

6. Evaluate Model – Measure accuracy using **Precision, Recall, and F1-score.**

7. Deploy Model – Automatically **classify incoming emails as spam or not spam.**

===

A **business wants to analyze customer reviews** of its products and determine whether the **sentiment is positive or negative**.
Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Identify the problem type: **Classification**

Step-by-step logic:

1. Collect Data – **Gather labeled customer reviews** (positive/negative).

2. Preprocess Text Data – **Remove stopwords, punctuation**, and **tokenize words**.

3. Convert Text into Features – Use TF-IDF or Word2Vec to **convert text into numerical format.**

4. Split Dataset – **Train-test split**.

5. Choose Algorithm – Use **Logistic Regression, Naive Bayes, or Transformers (BERT)**.

6. Train Model – **Fit the model on the training dataset.**

7. Evaluate Model – Use accuracy and **F1-score** to assess model performance.

8. Make Predictions – **Classify new customer reviews as positive or negative**.

===

An **insurance company** wants to predict whether a customer is likely to file a **claim in the next year** based on their **driving history and demographics**.
Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Identify the problem type: **Classification**

Step-by-step logic:

1. Collect Data – **Gather past claim history, driving behavior**, and customer demographics.

2. Preprocess Data – **Handle missing values** and encode categorical features.

3. Split Dataset – **Divide data into training and testing sets**.

4. Choose Algorithm – Use **Logistic Regression, Decision Tree, or Neural Networks.**

5. Train the Model – **Fit the model** using past claims data.

6. Evaluate Model – Use **Precision-Recall, AUC-ROC score**.

7. Deploy Model – **Predict claims likelihood for new customers**.

===

A **streaming platform** wants to **recommend movies to users** by grouping them **based on their viewing preferences** and **watch history**.
Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Identify the problem type: **Clustering**

Step-by-step logic:

1. Collect Data – **Gather user movie preferences**, genres watched, and **ratings**.

2. Preprocess Data – **Convert categorical** movie genres i**nto numerical format**.

3. Choose Clustering Algorithm – Use **K-Means or Hierarchical Clustering**.

4. Determine Optimal Clusters – Use the **Elbow Method**.

5. Train Model – **Apply clustering algorithm to group users.**

6. Analyze Clusters – **Identify user categories** (e.g., "**Action Lovers**,"
"**Drama Fans**").

7. Recommend Content – **Suggest movies based on cluster preferences**

==

A **hospital wants to predict the recovery time of patients** after surgery based on their age, medical history, and lifestyle habits.
Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Identify the problem type: **Regression**

Step-by-step logic:

1. Collect Data – **Gather historical recovery data** with features like patient **age, medical history, and lifestyle habits.**

2. Preprocess Data – **Normalize medical features** and handle missing values.

3. Choose Regression Algorithm – Use **Random Forest Regression or Linear Regression.**

4. Train Model – **Fit the model on training data**.

5. Evaluate Model – Use **RMSE (Root Mean Square Error)** to check accuracy.

6. Make Predictions – **Predict recovery time for new patients based on medical records**.

==

A **university wants to predict a student's final exam score** based on **study hours, attendance**, and past academic performance.
Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Identify the problem type: **Regression**

Step-by-step logic:

1. Collect Data – **Gather historical student records** with study hours, attendance, and exam scores.

2. Preprocess Data – **Handle missing values** and **standardize numerical features.**

3. Split Dataset – **Divide data into training and testing sets**.

4. Choose Algorithm – Use **Linear Regression or Support Vector Regression**.

5. Train the Model – **Fit the model on training data.**

6. Evaluate Performance – Use metrics like **RMSE and R^2 score.**

7. Make Predictions – **Estimate exam scores for new students based on input features**.

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